APEX TUITIONS

Daily Worksheet Series

Grade 8 • Mathematics

CBSE / State Board Aligned

Student Name: _____	**Roll No / Sec:** _____	**Date:** _____

Max Marks: 25	**Time Allowed:** 40 Mins	**Total Questions:** 11

DAILY PRACTICE WORKSHEET
GRADE 8 — MATHEMATICS
Topic: Linear Equations in One Variable

Name: _____
Roll No.: _____
Date: _____

SECTION A: Conceptual Recall & Objectives

1. The highest power (degree) of the variable in a linear equation is _____.

2. Which of the following represents a linear equation in one variable?
a) $x^2 + 2x = 5$
b) $3x + 2y = 8$
c) $4x - 7 = 13$
d) $x(x - 3) = 0$

3. State True or False:
In a linear equation, the transposed term changes its operational sign when moved from one side to the other.
Answer: [_____]

4. If $\frac{x}{4} = 3$, then the value of $3x - 5$ is equal to _____.

5. A linear equation in one variable has:
a) No solution
b) A unique solution
c) Two solutions
d) Infinitely many solutions

SECTION B: Short Practice Problems

(Show your working clearly in the spaces provided)

6. Solve for x:
$$5x - 9 = 2x + 6$$

\ \ \ \ **Answer:** $x =$ _____

7. Solve the following equation:
$$\frac{3y + 4}{2 - 6y} = -\frac{2}{5}$$

APEX TUITIONS

\ \ \ \ **Answer:** $y =$ _____

8. The sum of three consecutive integers is 72. Find the integers.

\ \ \ \ **Answer:** Integers are: _____, _____, _____

9. Solve for m:
$$m - \frac{m - 1}{2} = 1 - \frac{m - 2}{3}$$

\ \ \ \ **Answer:** $m =$ _____

SECTION C: Application & Critical Thinking

10. The present age of a father is three times the age of his son. Six years ago, the father's age was four times the age of his son. Formulate the linear equation and determine their present ages.

\ \ \ \ \ \ **Son's Age:** _____ years | **Father's Age:** _____ years

11. The perimeter of a rectangular swimming pool is 154 m. Its length is 2 m more than twice its breadth.
a) Frame the equation representing this condition.
b) Find the length and breadth of the pool.

\ \ \ \ \ \ **Length:** _____ m | **Breadth:** _____ m

APEX TUITIONS